

MLFF Architecture Revision 75

TARGET-DATA2C-MVMIGRATE1 atomic generated-policy migration latch

mdstats project

2026-08-16

Revision 75

Revision 75 implements TARGET-DATA2C-MVMIGRATE1 as the only legal transition from the revision-64 v4 dynamic-rescue path to the exact sparse multi-view fixed-eight generation. The migrated record family is TARGET-DATA2C v5, TARGET-DATA2D v3, and TARGET-DATA2E v3. v5 uses only REPAIR1 master-order prefixes at 128, 256, 512, 1024, 2048, 4096, 8192, and 16384; coverage and mandatory obligations are independently reconstructed rather than trusted from selector-internal scores; four hard qualifiers are required; and dynamic rescue is absent.

The implementation is deliberately two-phase because positive GPU qualification was frozen for final release. CPU/control-plane preparation may build and authenticate the exact v5 candidate and the migration decision lineage, but it cannot activate generated defaults. Activation requires passed MVQUAL1 paired legacy-vs-MV learning controls and a passed SIZE-FIDELITY2 report whose GPU status is exactly passed. Missing evidence remains pending; failure blocks migration. This preserves the project rule that GPU qualification occurs once against the complete release package.

Historical v4/v2/v2 records remain readable. v5/v3/v3 schemas are generation-separated so old authorities cannot masquerade as the migrated generation. Restart invalidation is content-addressed at the migration boundary, allowing valid DATA6, TARGET-DATA2A/B, MVIDX1, MVSEL1, and REPAIR1 products to survive final-GPU evidence updates. DATA8 membership, e3nn source/DATA6 policy, CuEq TRAIN2 policy, and the 0.95 independent hard-coverage threshold are unchanged.

Release: mdstats 0.20.208a0

Dependency-graph schema: 57

Next action: FINAL-GPU1 consolidated qualification and atomic v5/v3/v3 activation on pass.